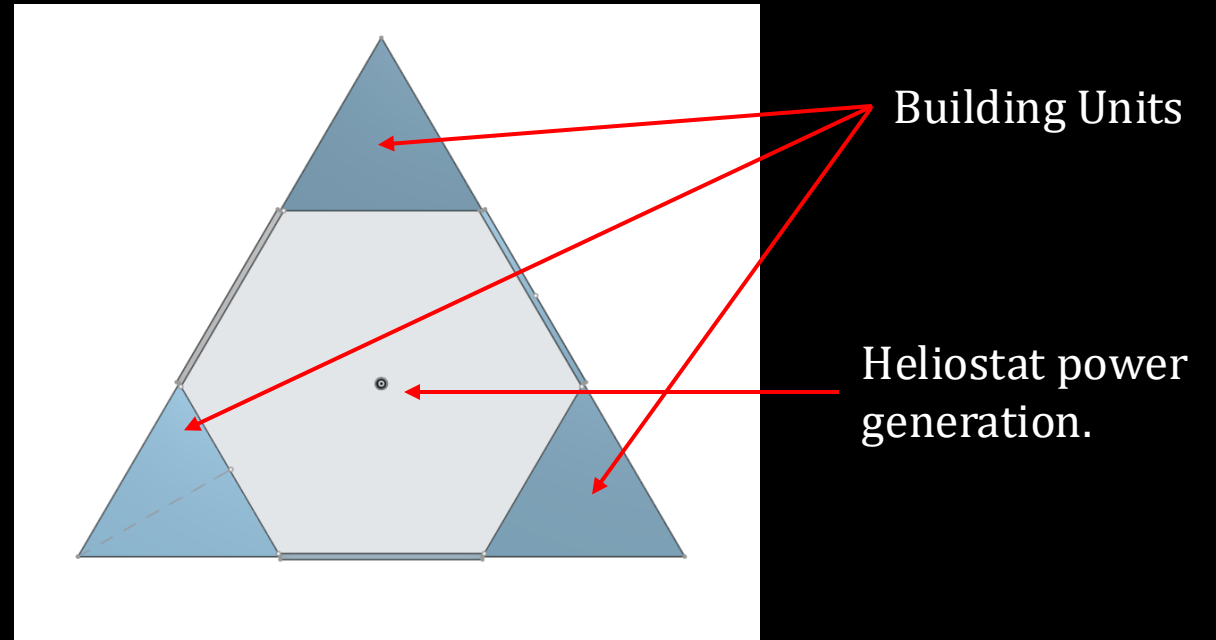
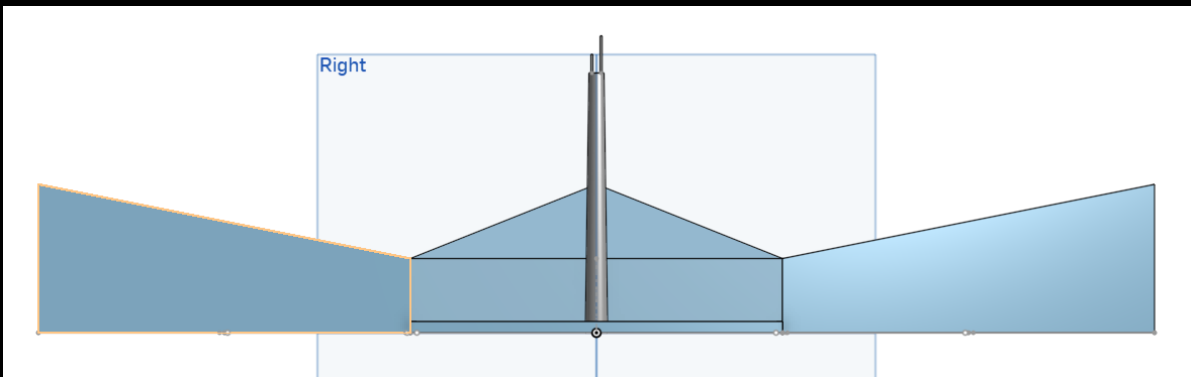
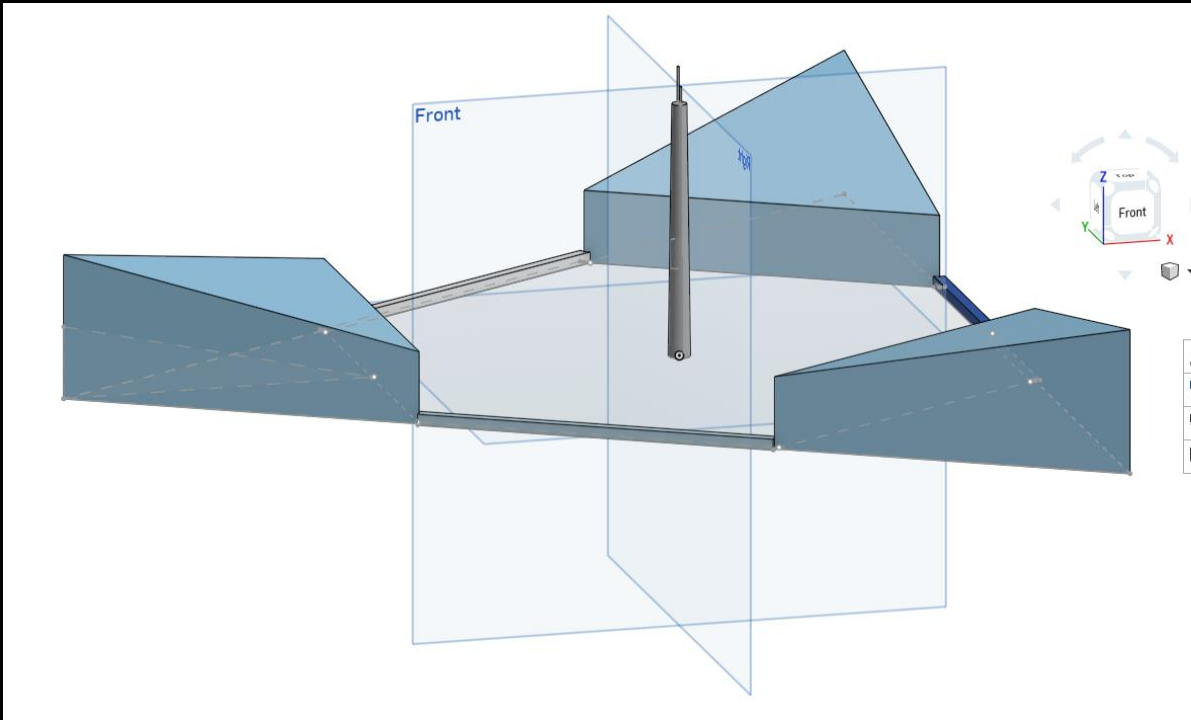




Boscom

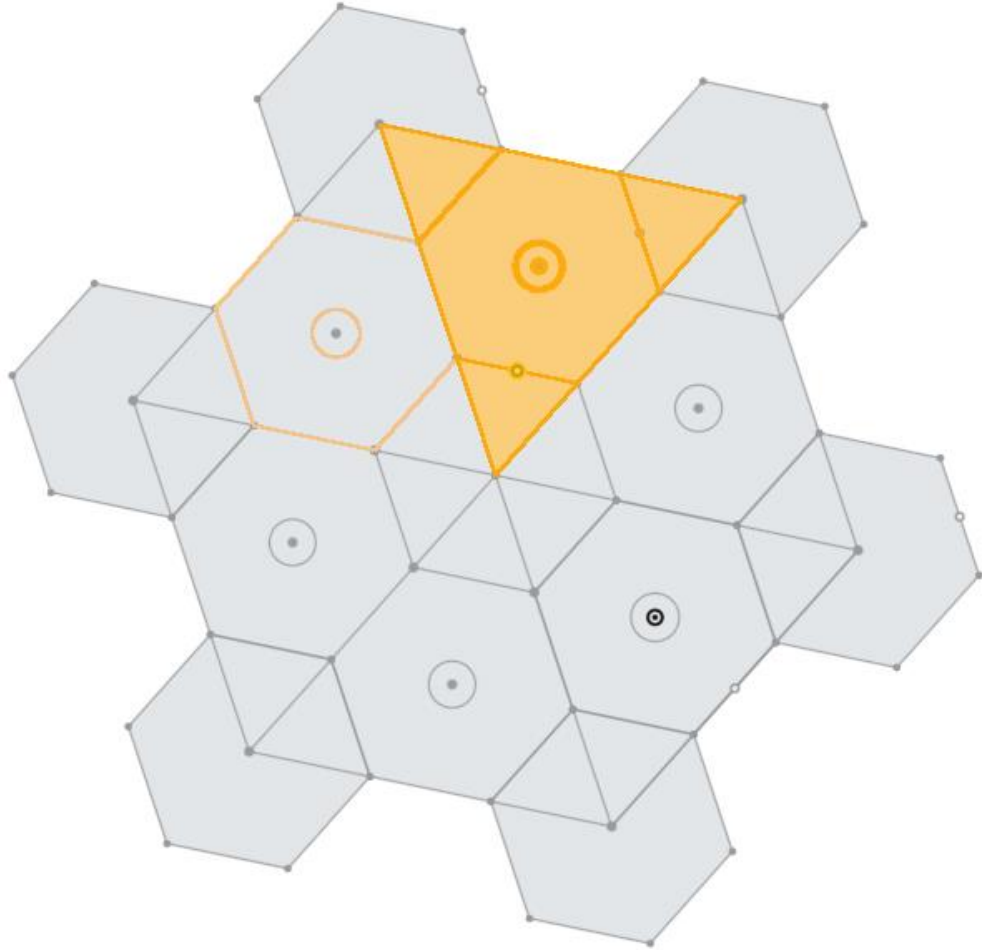
Poynting Crater

Triangular Base Unit(S1)



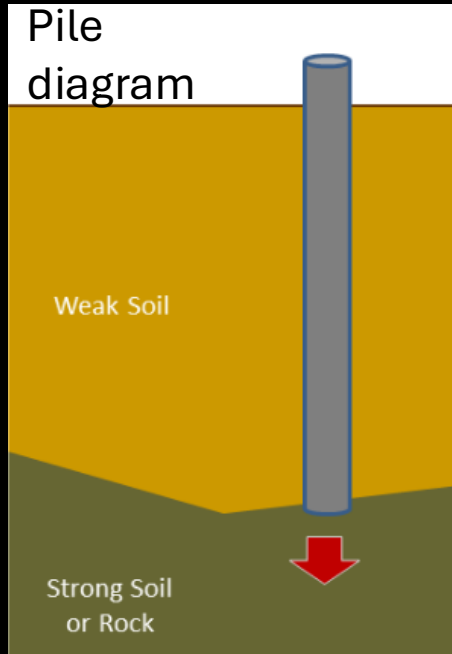
The above-mentioned building units will have there purpose (Farm and storage space, Manufacturing facilities and Residential spaces) will be delegated as demand requires.

Modular Expansion(S5)



As shown in the diagram to the right, the base unit seen on Slide 2 can be configured to resemble a hexagonal, crystalline structure. In this analogy, the diagram to the right would be the crystal for all the rest of the base to expand upon. Furthermore, I have chosen to do the hexagonal structure in order to maximise the use of space as hexagons can perfectly tessellate.

Materials(S3 and S6)



Regolith and sulfur are used to produce Martian based concrete which can be used to construct a solid base (including piles to stabilise the structures on potentially dusty and loose terrain) and the other walls and roofs of the structure. Martian regolith concrete is particularly good due to its availability, immense strength, durability, low cost, and resistance to fire and radiation due to its high density of $2.2\text{--}2.4\text{g/cm}^3$.

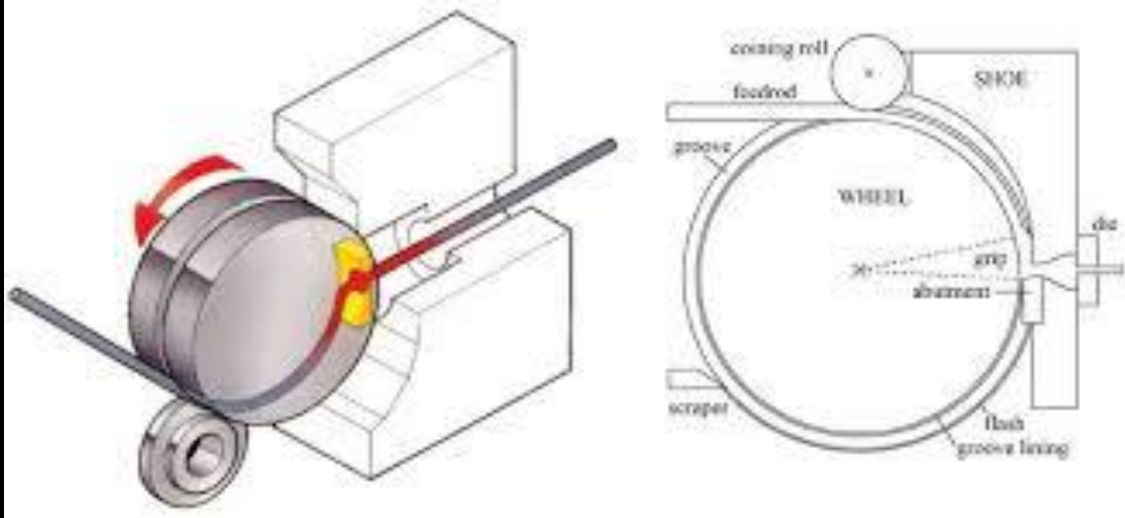


Fused Silica Windows can be used which utilize self-healing acrylic for impact recovery, retain 1.5 atm of pressure at a thickness of 10mm and span of 2m.



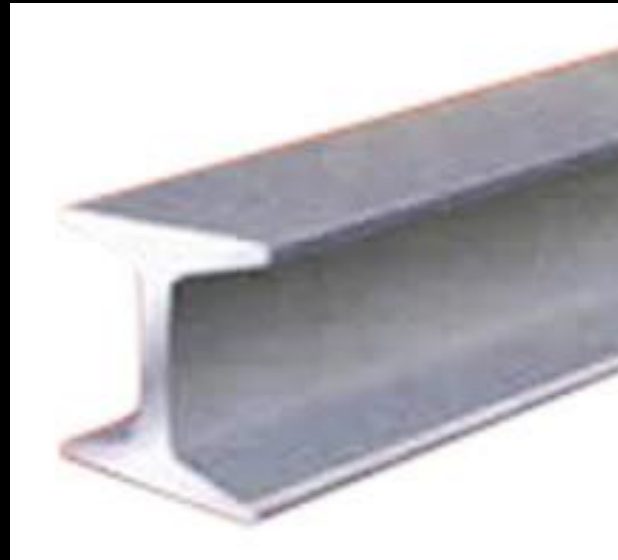
Exposed surfaces such as Blinds can be made of Polyethylene (a common plastic used in space craft and spacesuits due to its hydrogen rich content which is useful as high ratio of electrons to nucleons and can break down high-energy particles without creating excessive secondary radiation).

Production(M1)

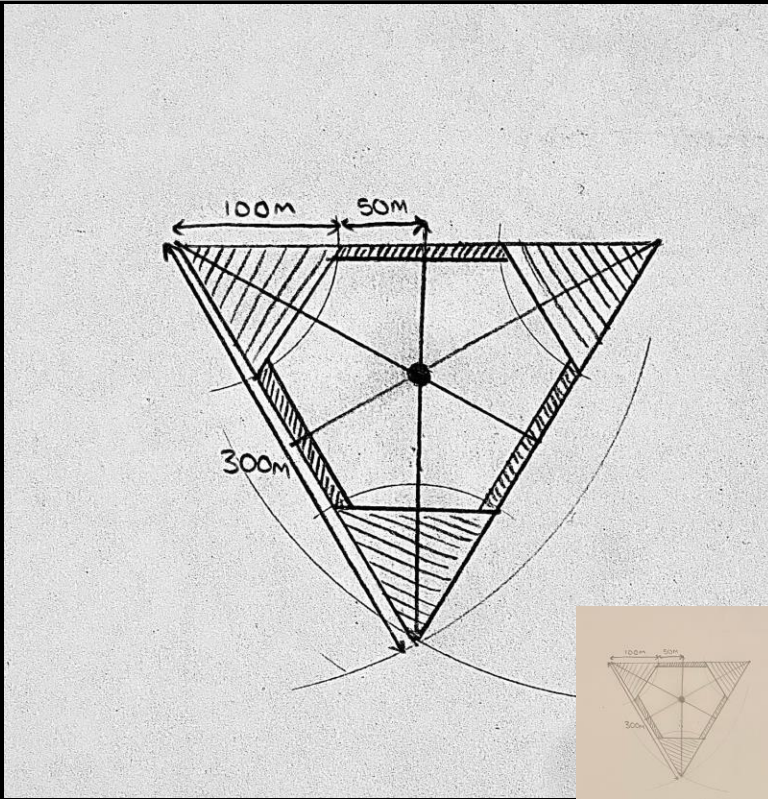


Supplied Iron can be extruded into slabs and I beams. I chose extrusion over other processes like hot pressing or die casting as extrusion is a form of continuous production and thus also makes the produced slab or beam be of any desired length. This process also could be used to make the walls of a CASSSC. For more unusually shaped or bespoke products, die casting will be used to create accurate and detailed replicas.

CASSSCs



Power Supply(O4)



Heliostats can be used to take advantage of the considerably thinner of Martian atmosphere (about 100 times less dense than Earth's) which leads to an increase in solar energy. Additionally, as the power supply is set into the modular design, the surrounding manufacturing plants and lighting for housing and farms is easily and readily available nearby.

The approximate area of the shaded region(the area most suitable for heliostats) can be calculated as follows:

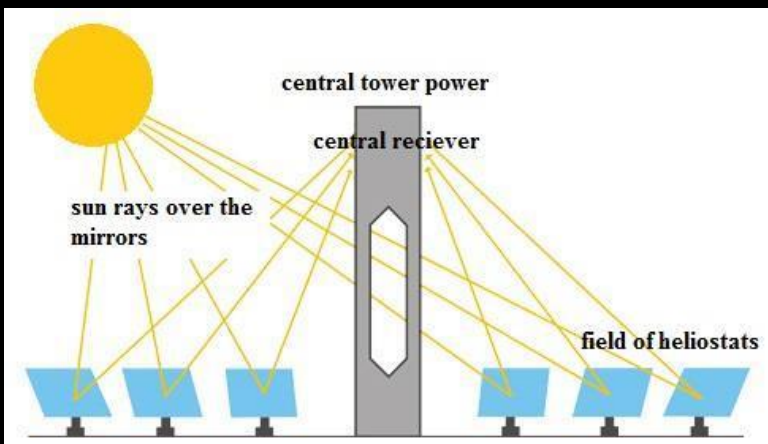
$$3(100 \times 3) + 3(100 \times \frac{1}{2} \times a)$$

Where $a = \sqrt{100^2 - 50^2}$

And therefore, the area = $900 + 12990.38 = 13890 \text{m}^2$ (to 4sf)

$13890 \times 2.0 \times 10^{10}$ (the energy provided per day per m^2 in joules) = 2.78×10^{14}

Excess energy is stored in large batteries for later use including in the event of solar coverage.



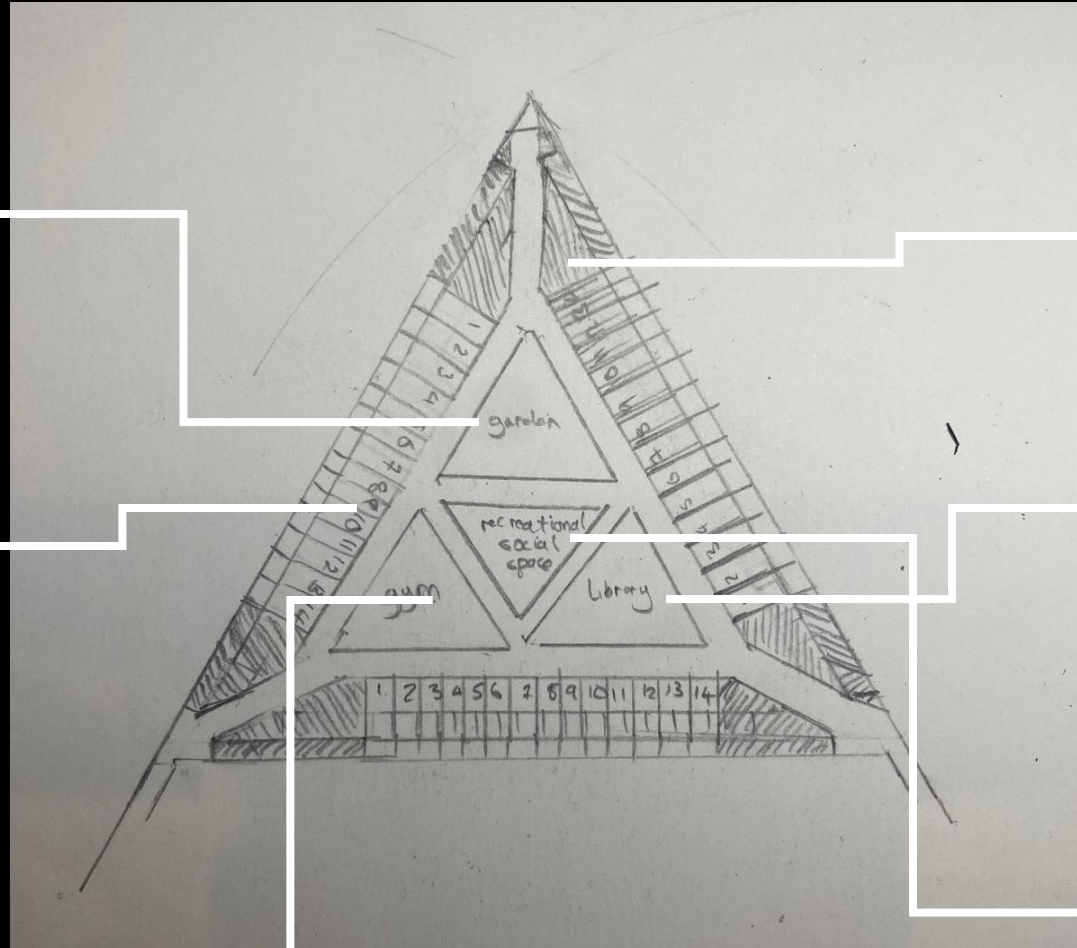
Furthermore, there will be antenna-like structures on the top of the central receiver (as demonstrated by the small pillars seen on the central receiver on Slide 2) can be used as points of elevation for enhanced satellite and extraplanetary communications.

Human Space (H1 and H2)

An Artificial garden is implemented to increase human mental health.

Numbered boxes represent rooms which have approximate sizes of 44m^2 , and 2 layers of this design results in rooms being shared by approximately 4 people

Gym can be used to maintain fitness and Human mental health.



Shaded areas represent community kitchen, dining areas and toilets and a staircase to higher levels.

Library space can be used to work, relax, research and increase intellect

Recreational space offers residents a place to relax and socialize thus boosting human morale and mental health

Necessary Imports

Fertiliser

Alfalfa seeds should be imported as they can grow in Martian soil and their biomass can be used to fertilise the soil.

Oxygen and Water

Nitrogen-Hydro ships nitrogen, oxygen, and clean water from its facilities. Liquid nitrogen and water are shipped in sealed CASSC's from their production facilities at Alaskol and Argonom.

Liquid nitrogen is priced at \$0.15 per litre, liquid oxygen at \$1.5 per litre, while the flagship product, water, is priced \$5 per litre.

These resources will be used to sustain human life.

Farming

For the farming building unit, I will be using vertical farming in order to be more space efficient for the building spaces. Plants such as Potatoes and leafy greens -such as spinach or kale- should be grown due to their high yield, nutritional value and resilience. Additionally, the base will utilise hydroponics to further increase the yield.



Thank You For Watching!

I would like a role in either: Structure, Operations or Missions as they are physics based and my interests in further education reside in mathematics, physics and chemistry.

3d and 2d models and sketches were made by me. Attached in the bottom right is an RC plane which I made recently because like this competition, physics and engineering is something I find genuinely fun and I would love to be a part of

